ModelIt!

Student Activity Pack

Why Do Animals Migrate Thousands of Miles?

Modeling the Costs and Benefits of Long-Distance Animal Migration

Grade 10L2 | NGSS HS-LS2-8, HS-LS4-6 | Cause and Effect

STUDENT NAME _____ DATE _____

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Table of Contents

ModelIt!

1 Pre-Assessment 3

2 Scoring Rubric 4

3 Component Analysis 5

4 Model Recording Page 6

5 Simulation Observations 7

6 Research & Extend 8

7 Reflection Questions 9

8 STEM Challenge 10

1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **Why do you think animals like birds, whales, and butterflies migrate thousands of miles instead of staying in one place?**

2. **How do you think animals navigate across oceans and continents without maps or GPS?**

3. **Draw a diagram showing what you think a migratory bird needs to survive a 5,000-mile journey.**

4. **What threats do you think migratory animals face today that they didn't face 100 years ago?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Navigation Accuracy | Energy Reserves | Climate Trigger | Predator Avoidance | Group Size | Food Availability | Reproductive Success

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Optimal Conditions

Aligned timing, abundant stopovers, large groups

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Climate Mismatch

Food peaks earlier than arrival time

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Stopover Loss

Critical refueling sites destroyed

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Light Pollution — Artificial light from cities that disorients nocturnal migrating birds, causing them to circle illuminated buildings until exhaustion — an estimated 1 billion birds die annually from building collisions during migration in the US alone

Wind Pattern Shifts — Changes in prevailing wind patterns due to climate change that alter the energy cost of migration — tailwinds become headwinds, increasing the fat reserves needed and the risk of exhaustion during ocean crossings

Genetic Program Fidelity — The accuracy of the inherited genetic instructions that encode migration direction, distance, and timing — which can only evolve over many generations and cannot adapt to rapid environmental changes within a few decades

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Why would an animal risk a journey of thousands of miles through storms, predators, and starvation — and how do millions of them navigate with pinpoint accuracy? Use evidence from your simulation to support your answers.

1. **How does your model demonstrate that migration is an evolutionary cost-benefit calculation rather than a simple instinct?**

2. **What does your model reveal about why phenological mismatch is a more serious threat than direct habitat loss for some migratory species?**

3. **How does Group Size improve individual survival — what mechanisms make collective migration safer than solo migration?**

4. **Why are stopover sites the 'weakest links' in the migration system, according to your model?**

5. **What are the limitations of modeling migration with only seven components when real migration involves weather unpredictability, individual learning, and complex social dynamics?**

8 STEM Challenge: Design a Migration Corridor Conservation Plan

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YOUR ENGINEERING MISSION

Design a conservation strategy that protects a complete migration corridor — including breeding grounds, stopover sites, and wintering habitat — for a threatened migratory species.

THE SCENARIO

A conservation organization has identified that a migratory shorebird population has declined 70% in 20 years. Analysis shows the decline is caused by loss of three critical stopover sites along their 10,000 km migration route. Your team has been hired to design a corridor conservation plan that protects the most critical habitats across multiple countries.

Engineering Design Thinking

1. WHAT CRITERIA WOULD YOU USE TO PRIORITIZE HABITAT PROTECTION — STOPOVER SITE QUALITY, THREATENED SPECIES DENSITY, FEASIBILITY OF PROTECTION, OR COST?

2. HOW WOULD YOU DESIGN PROTECTED AREAS THAT ARE RESILIENT TO CLIMATE CHANGE SHIFTING THE MIGRATION TIMING?

3. WHAT ECONOMIC INCENTIVES COULD ENCOURAGE LOCAL COMMUNITIES ALONG THE ROUTE TO PROTECT MIGRATION HABITAT?

4. HOW WOULD YOU USE SATELLITE TRACKING AND POPULATION DATA TO MEASURE CONSERVATION OUTCOMES?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Migration Ecologists study the movement patterns, physiology, and conservation needs of migratory animals. They work for wildlife agencies, conservation organizations, and universities, earning $55,000–$120,000/year. Wildlife Corridor Planners who design connected habitat networks earn $60,000–$110,000/year.